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Week3_Capstone_SOC_Workflow_Simulation

1. Introduction

This capstone project was designed to simulate a complete **Security Operations Center** workflow, from attack execution to detection, triage, response, escalation, and reporting. The purpose of the exercise was to connect multiple SOC activities into one incident-handling process instead of treating them as isolated tasks.

In this lab, a Metasploit attack was launched from Kali Linux against a vulnerable Metasploitable2 target using the Samba usermap script exploit. Because Wazuh was installed on the Ubuntu monitoring server and not directly on Metasploitable2, the exploit itself was not directly detected by Wazuh. Instead, the workflow used the same attacker IP to generate a Wazuh-visible suspicious event on the monitored Ubuntu host, which allowed detection, triage, and response stages to be completed consistently within the lab.

The capstone therefore demonstrated a realistic SOC workflow with one important limitation: direct exploit telemetry was not available on the vulnerable target, so detection and containment were correlated through attacker IP activity observed on the monitored environment.

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2. Objectives

The objectives of this capstone exercise were to:

- simulate a real attack using Metasploit in a controlled lab;
- identify and document attacker activity in Wazuh;
- triage the suspicious event and classify its severity;
- apply a response action through CrowdSec;

- escalate the incident to Tier 2;
- produce a technical incident report and a non-technical management briefing.

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3. Tools Used

The following tools were used in this exercise:

- **Metasploit** for attack simulation;
- **Wazuh** for detection and triage;
- **CrowdSec** for containment decision-making;
- **TheHive case draft workflow** for escalation;
- **Google Docs / Word** for formal reporting and briefing.

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4. Incident Scenario

The attack simulation was executed from a Kali Linux attacker machine against a vulnerable Metasploitable2 target. The exploit used in Metasploit was the Samba usermap script module, which represents exploitation of a remote service. After the exploit was successfully launched, the same attacker IP was reused to generate authentication activity visible to the monitored Ubuntu system, allowing the SOC workflow to continue inside Wazuh.

Lab Mapping

Role	System
Attacker	Kali Linux
Vulnerable target	Metasploitable2
Monitoring and response platform	Ubuntu with Wazuh and CrowdSec
Reported server name for assignment consistency	Server-Y

5. Attack Simulation

The attack simulation was carried out in Metasploit using a Samba vulnerability on Metasploitable2. The objective of this step was to establish the offensive phase of the incident and identify the attacker source IP that would later be used in detection and response.

Attack Details

Field	Value
Attack Platform	Kali Linux
Target	Metasploitable2
Exploit Module	exploit/multi/samba/usermap_script
Attack Type	Remote service exploitation
MITRE ATT&CK	T1210 – Exploitation of Remote Services
Result	Exploit successfully executed

The Metasploit stage confirmed that hostile activity originated from the attacker IP and that the exploit phase of the capstone was completed successfully.

6. Detection and Triage

Because Wazuh was not directly monitoring the Metasploitable2 machine, the next stage of the workflow focused on attacker IP activity visible on the monitored Ubuntu host. The same attacker IP generated suspicious authentication behaviour that was then reviewed in Wazuh.

Detection Record

Timestamp	Source IP	Alert Description	MITRE Technique
Mar 29, 2026 @ 00:53:29.237	192.168.56.102	Threat intelligence match: source IP found in AlienVault reputation feed / failed SSH authentication activity	T1210

Supporting Alert Details

- **Host:** Hasina
- **Affected user:** hasina
- **Source IP:** 192.168.56.102
- **Source port:** 54482
- **Rule ID:** 100200
- **Rule Level:** 10
- **Rule Groups:** threat_intel, otx_enriched, ip_reputation, authentication_failed
- **Original log:** Failed password for hasina from 192.168.56.102 port 54482 ssh2

Triage Assessment

The event was triaged as **High priority** because:

- the source IP was the same IP used in the exploit simulation;
- the IP matched the imported AlienVault reputation feed in the lab;
- the event involved authentication failure behaviour against a monitored system;
- the attacker activity was part of a broader exploitation scenario.

Triage Decision

Priority	Status	Analyst Decision
High	Open / Escalated	Treat as suspicious attacker activity linked to exploitation scenario

7. Response and Containment

CrowdSec was used as the response platform for containment. A manual decision was created against the attacker IP in order to represent the blocking action required by the capstone scenario.

Containment Action

Action	Target	Result
CrowdSec decision created	192.168.56.102	Decision successfully added
Verification	Repeat SSH attempt / connectivity test	Access still possible

Containment Interpretation

The CrowdSec decision was created successfully, but the attacker IP was not fully blocked at the firewall layer because no active CrowdSec bouncer was registered at the time of testing. As a result, the containment stage was documented as a remediation action rather than a fully enforced network block.

This was an important operational lesson: creating a decision in CrowdSec is not sufficient by itself. A bouncer or remediation component must also be active in order for the block to be applied to the system firewall.

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8. Escalation to Tier 2

After triage and initial response, the incident was escalated to Tier 2 for deeper review.

Tier 2 Case Summary

A high-priority security incident was identified during a lab capstone exercise involving exploitation activity from source IP 192.168.56.102. A Metasploit attack simulation successfully targeted a vulnerable Metasploitable2 service consistent with **MITRE ATT&CK T1210.**

Follow-on suspicious activity from the same source IP was detected and reviewed in Wazuh, then addressed through a CrowdSec containment decision.

Because the attacker IP was associated with exploitation-related behaviour and the event required further validation of scope and impact, the incident was escalated from Tier 1 to Tier 2. Further investigation is required to confirm the full timeline, correlate attack stages, and verify whether additional assets were exposed.

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9. Situation Report (SITREP)

Incident Title: Unauthorized Access / Exploitation Activity on Server-Y

Date/Time Detected:
Mar 29, 2026 @ 00:53:29.237

Source IP:
192.168.56.102

Summary:
A suspicious attacker IP associated with a prior exploitation simulation was identified in Wazuh through failed authentication activity on the monitored environment. The incident was treated as high priority because the source IP was linked to the offensive phase of the capstone and matched the threat intelligence enrichment workflow used in the lab.

Affected Asset:
Server-Y (assignment label) / monitored Ubuntu host

Actions Taken:
The alert was reviewed in Wazuh, triaged as high priority, documented, and addressed through a CrowdSec decision. The case was then escalated for deeper investigation.

Current Status:
Escalated to Tier 2 and documented for reporting.

Recommended Next Steps:
Validate the full attack timeline, improve direct telemetry on vulnerable targets, ensure CrowdSec bouncer enforcement is active, and preserve evidence related to the attacker IP and authentication events.

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10. Formal Incident Report

Executive Summary
A capstone SOC simulation was conducted to test attack execution, detection, triage, response, escalation, and reporting. A Metasploit exploit was successfully launched from Kali Linux against a vulnerable Metasploitable2 service using the Samba usermap script module.

The same attacker IP was later correlated with suspicious authentication activity visible on the monitored Ubuntu environment. Because the attacker IP was linked to the exploitation phase and triggered threat intelligence-related review in Wazuh, the incident was classified as high priority. CrowdSec was used to create a containment decision, and the incident was escalated for further investigation.

Timeline

- Metasploit exploit launched from Kali against Metasploitable2
- Exploit completed successfully
- Suspicious activity from attacker IP observed on monitored Ubuntu host
- Wazuh alert reviewed and triaged as high priority
- CrowdSec decision added for attacker IP
- Incident escalated to Tier 2
- Reporting and management briefing completed

Recommendations

- Add direct telemetry collection on vulnerable target systems for stronger exploit detection;
- maintain attacker IP correlation between attack, detection, and response stages;
- enable an active CrowdSec bouncer to enforce decisions at the firewall layer;
- standardize escalation and SITREP templates;
- continue running end-to-end SOC simulations to improve workflow readiness.

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11. Non-Technical Manager Briefing

A controlled cyberattack simulation was performed in the lab to test how the security workflow responds to a real incident. A vulnerable service was successfully targeted from an attacker machine, and the same source IP was later identified and reviewed through the monitoring process.

The event was treated as high priority, documented, and addressed through a CrowdSec containment decision before being escalated for deeper investigation. The exercise showed that the team can connect attack simulation, monitoring, response, and reporting into one structured workflow, while also revealing that stronger direct monitoring of vulnerable targets and full enforcement of containment actions are still needed.

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12. Findings Summary

The capstone successfully demonstrated a full SOC workflow from attack simulation to reporting. Metasploit provided the offensive stage, Wazuh supported detection and triage through attacker IP activity, CrowdSec represented the containment decision, and the incident was escalated and documented through formal reporting. The main limitation was the absence of direct exploit telemetry on the target system.

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13. Conclusion

This capstone project successfully connected multiple SOC stages into one coherent incident workflow. A real exploit was simulated against Metasploitable2, the attacker IP was tracked and reviewed through Wazuh, a response decision was created in CrowdSec, the incident was escalated to Tier 2, and the final findings were documented in both technical and management-facing formats.

The exercise also highlighted an important operational lesson: effective SOC response depends not only on having multiple tools, but on understanding exactly where visibility exists and where it does not. In this lab, Wazuh did not directly detect the exploit on the vulnerable target, but it still played an important role in correlating attacker behavior on the monitored environment. CrowdSec supported the response stage, although full enforcement required an active bouncer. Overall, the capstone demonstrated practical SOC reasoning, structured incident handling, and realistic reporting.

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14. Screenshot References

- **07_Metasploit_Attack_Simulation.png** — Successful Metasploit exploit execution against Metasploitable2
- **08_Wazuh_Detection.png** — Wazuh detection tied to the attacker IP
- **09_CrowdSec_Block_Action.png** — CrowdSec decision used for containment